Beyond Spherical Symmetry: A Weakened Condition Toward Feasible Universal Feature Selection

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1. Introduction

- Feature selection [1] is crucial in high-dimensional learning and inference.
- While task-specific features can be obtained with sufficient domain knowledge, a more ambitious goal is to extract **universally effective** features that remain applicable across a diverse range of tasks.
- We operates within the Markov chain framework $U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$, focusing on the use of features derived from the observed variables X and Y to infer the latent attributes U and V.

2. Preliminaries

- ϵ-neighborhood: $\mathcal{N}_{\mathcal{Z}}^{\epsilon}(P_Z) = \{P \in \mathcal{P}^{\mathcal{Z}} : \mathcal{X}^2(P \| P_Z) \le \epsilon^2\}$.
- ϵ-attribute: W is an ϵ-attribute of Z, if $P_{Z|W}(\cdot \mid w) \in \mathcal{N}_{\mathcal{Z}}^{\epsilon}(P_Z)$ for any $w \in \mathcal{W}$. Its configuration $\mathcal{C}_{\mathcal{Z}}^{\epsilon}(P_Z) = \{\mathcal{W}, \{P_W(w)\}, \{P_{Z|W}(\cdot \mid w)\}\}$.
- Information matrix & vector: For $z_i \in \mathcal{Z}$, $w_j \in \mathcal{W}$,

$$\boldsymbol{\Phi}_{i,j}^{Z|W} = \frac{1}{\epsilon} \cdot \frac{P_{Z|W}(z_i \mid w_j) - P_Z(z_i)}{\sqrt{P_Z(z_i)}}.$$

  The j-th column $\phi_{w_j}^{Z|W}$ is the information vector for w_j.
- Feature vectors: Given normalized features $h^k = (h_1, \cdots, h_k)^{\mathrm{T}} : \mathcal{Z} \to \mathbb{R}^k$, define $\psi_i^Z(z) = \sqrt{P_Z(z)}\, h_i(z), i = 1, \ldots, k$.
- Canonical dependence matrix: For X, Y, let $K_{XY} = \min\{|\mathcal{X}|, |\mathcal{Y}|\}$ and

$$(\tilde{\boldsymbol{B}}_{X,Y})_{j,i} = \frac{P_{X,Y}(x_i, y_j) - P_X(x_i) P_Y(y_j)}{\sqrt{P_X(x_i)}\sqrt{P_Y(y_j)}}.$$

3. Problem Setup

- Markov chain $U \overset{\epsilon}{\leftrightarrow} X \leftrightarrow Y \overset{\epsilon}{\leftrightarrow} V$, where U, V are the ϵ-attribute of X, Y.
- Data $x^N = (x_1, \cdots, x_N)$, $y^N = (y_1, \cdots, y_N)$ i.i.d. drawn from $P_{X,Y}$.
- Statistics S^k and T^k are the empirical expectations of $f^k(X)$ and $g^k(Y)$.
- The collection of configurations $\mathcal{C}_{\mathcal{X}}^{\epsilon}(P_X)$ and $\mathcal{C}_{\mathcal{Y}}^{\epsilon}(P_Y)$ are equipped with probability measures μ_U and μ_V.
- The error probability for inferring U from S^k is $p_e^{U|S}(f^k, \mathcal{C}_{\mathcal{X}}^{\epsilon}(P_X))$, and the error exponent

$$\bar{E}^{U|S}(f^k) = \lim_{N\to\infty} -\frac{\mathbb{E}_{\mu_U}\left[\log p_e^{U|S}(f^k, \mathcal{C}_{\mathcal{X}}^{\epsilon}(P_X))\right]}{N}$$

  Similarly define $\bar{E}^{V|S}(f^k)$, $\bar{E}^{U|T}(g^k)$ and $\bar{E}^{V|T}(g^k)$.
- We aim to seek features f^k and g^k that maximize the above exponents.

4. Previous Results

Recent works [2, The 1] and [3, Prop 5.10] assume μ_U and μ_V exhibit **no directional preference**, i.e. $\Phi^{X|U}$ and $\Phi^{Y|V}$ satisfy spherical symmetry [4].

Definition 1 (Spherical symmetry) *A random matrix $\boldsymbol{A} \in \mathbb{R}^{n\times m}$ is spherically symmetric, if for any orthogonal matrices $\boldsymbol{Q}_1 \in \mathbb{R}^{n\times n}$ and $\boldsymbol{Q}_2 \in \mathbb{R}^{m\times m}$, we have $\boldsymbol{Q}_1^{\mathrm{T}} \boldsymbol{A} \boldsymbol{Q}_2 \overset{d}{=} \boldsymbol{A}$.*

Theorem 1 (Huang et al. 2024) *For $k \in \{1, \ldots, K_{XY} - 1\}$, select normalized features $f^k : \mathcal{X} \to \mathbb{R}^k$ and $g^k : \mathcal{Y} \to \mathbb{R}^k$. Assume under each probability measures, both $\boldsymbol{\Phi}^{X|U}$ and $\boldsymbol{\Phi}^{Y|V}$ are spherically symmetric. Then,*

$$\begin{aligned}&\left(\bar{E}^{U|\hat{S}}(f^k), \bar{E}^{V|\hat{S}}(f^k), \bar{E}^{U|\hat{T}}(g^k), \bar{E}^{V|\hat{T}}(g^k)\right) \le \\ &\left(C_U \epsilon^2 k,\ C_V \epsilon^2 \sum_{i=1}^{k} \sigma_i^2,\ C_U \epsilon^2 \sum_{i=1}^{k} \sigma_i^2,\ C_V \epsilon^2 k\right) + \mathrm{o}(\epsilon^2),\end{aligned} \tag{1}$$

as $\epsilon \to 0$, where C_U and C_V are positive constants independent of ϵ, k or $P_{X,Y}$, and σ_i is the i-th largest singular value of $\tilde{\boldsymbol{B}}_{X,Y}$. Moreover, the equality holds when the feature vectors corresponding to f_i and g_i are precisely the right and left singular vectors of $\tilde{\boldsymbol{B}}_{X,Y}$ corresponding to σ_i.

5. Motivation and Observation

The exact spherical symmetry is rather restrictive and challenging to achieve:

Lemma 2 *If $\boldsymbol{A}$ is spherically symmetric, then, (1) $\mathbb{E}[\boldsymbol{A}] = \boldsymbol{0}_{n\times m}$; (2) For any i, j, k, l, $\boldsymbol{A}_{i,j} \overset{d}{=} \boldsymbol{A}_{k,l}$; (3) For $(i,j) \neq (k,l)$, $\mathrm{Cov}(\boldsymbol{A}_{i,j}, \boldsymbol{A}_{k,l}) = 0$.*

We have the following observation. Assume W is an ϵ-attribute of Z, then

$$\log \frac{P_{Z|W}(z \mid w)}{P_Z(z)} = \epsilon \frac{\phi_w(z)}{\sqrt{P_Z(z)}} - \frac{\epsilon^2}{2}\Big(\frac{\phi_w(z)}{\sqrt{P_Z(z)}}\Big)^2 + \mathrm{o}(\epsilon^2),$$

where ϕ_w is the information vector for w. For any smooth one-parameter family $\{w(\theta)\}$ passing through w at $\theta = 0$, the score function satisfies

$$\frac{\partial}{\partial\theta} \log P(z \mid w(\theta))\bigg|_{\theta=0} = \epsilon \frac{\phi_w'(z)}{\sqrt{P_Z(z)}} + \mathrm{O}(\epsilon^2),$$

where ϕ_w' denotes the derivative in the direction of θ. Then the Fisher information becomes $I(0) = \epsilon^2 \|\phi_w'\|^2 + \mathrm{O}(\epsilon^3)$. Hence, by the **Cramér–Rao lower bound**, as $\epsilon \to 0$, the term $\|\phi_w'\|^2$ dominates the mean-square error of estimation, indicating that in the local regime, imposing spherical symmetry can be reduced to a **second-moment symmetry condition**.

6. Our Main Contributions

We propose weak spherical symmetry based on second-moment deviation.

Definition 2 (Weak spherical symmetry) *For random matrices $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{R}^{n\times m}$, define a second-moment distance as*

$$\mathrm{D}(\boldsymbol{X}, \boldsymbol{Y}) = \sup_{u\in\mathbb{R}^n, v\in\mathbb{R}^m, \|u\|=\|v\|=1} \left|\mathbb{E}\big[(u^{\mathrm{T}}\boldsymbol{X}v)^2\big] - \mathbb{E}\big[(u^{\mathrm{T}}\boldsymbol{Y}v)^2\big]\right|.$$

Given $\delta \ge 0$, a random matrix $\boldsymbol{A} \in \mathbb{R}^{n\times m}$ is δ-spherically symmetric, *if for any orthogonal matrices $\boldsymbol{Q}_1 \in \mathbb{R}^{n\times n}$, $\boldsymbol{Q}_2 \in \mathbb{R}^{m\times m}$, we have $\mathrm{D}(\boldsymbol{Q}_1^{\mathrm{T}} \boldsymbol{A} \boldsymbol{Q}_2, \boldsymbol{A}) \le \delta$.*

Remark 1 *Spherical symmetry indicates δ-spherical symmetry with $\delta = 0$.*

Example 1 *Given $0 < |\sigma^2 - 1| \le \delta$. A random matrix $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ has independent entries $\boldsymbol{A}_{1,1} \sim N(0, \sigma^2)$ and $\boldsymbol{A}_{1,2}, \boldsymbol{A}_{2,1}, \boldsymbol{A}_{2,2} \sim N(0,1)$. Then, $\boldsymbol{A}$ is δ-spherically symmetric, but not exactly spherically symmetric.*

The weakened condition allows bounded ($\le \delta$) directional preference to exist. Under this condition, we demonstrate the following theorem.

Theorem 3 (Tang, Han, 2026) *For $k \in \{1, \ldots, K_{XY} - 1\}$, select normalized features $f^k : \mathcal{X} \to \mathbb{R}^k$ and $g^k : \mathcal{Y} \to \mathbb{R}^k$. Given $\delta \ge 0$, assume both $\boldsymbol{\Phi}^{X|U}$ and $\boldsymbol{\Phi}^{Y|V}$ are δ-spherically symmetric. Then,*

$$\begin{aligned}&\left(\bar{E}^{U|\hat{S}}(f^k), \bar{E}^{V|\hat{S}}(f^k), \bar{E}^{U|\hat{T}}(g^k), \bar{E}^{V|\hat{T}}(g^k)\right) \le \\ &\left(C_U \epsilon^2 k,\ C_V \epsilon^2 \sum_{i=1}^{k} \sigma_i^2,\ C_U \epsilon^2 \sum_{i=1}^{k} \sigma_i^2,\ C_V \epsilon^2 k\right) + \mathrm{O}(\delta\epsilon^2),\end{aligned} \tag{2}$$

as $\epsilon \to 0$, where C_U and C_V are positive constants independent of ϵ, k or $P_{X,Y}$, and σ_i is the i-th largest singular value of $\tilde{\boldsymbol{B}}_{X,Y}$. Moreover, the equality holds when the feature vectors corresponding to f_i and g_i are precisely the right and left singular vectors of $\tilde{\boldsymbol{B}}_{X,Y}$ corresponding to σ_i.

Remark 2 *Specifically, when $\delta = \mathrm{o}(1)$ as $\epsilon \to 0$, (2) reduces to (1), which implies the exact spherical symmetry condition is unnecessary in Theorem 1.*

7. Conclusion

- Theorem 3 recovers Theorem 1, which reveals the exact symmetry condition is unnecessary, and a small directional preference $\delta = \mathrm{o}(1)$ is tolerable.
- Even if δ is not relatively small (e.g., of order $\mathrm{O}(1)$), the proposed selection framework remains meaningful: the error bound is **governed by** and **robust against** the second-moment deviation δ.
- Our work extends the selection framework to a broad class of approximately symmetric environments. We believe it provides a robust, practical theory for universal feature selection in machine learning and statistical inference.

DETAILS AVAILABLE

D. Tang and G. Han, "Universal Feature Selection with Noisy Observations and Weak Symmetry Conditions," *arXiv preprint arXiv:2605.09396*, 2026.

KEY REFERENCES

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